First steps in Maestro

Learn how to set up Maestro projects, customize paths, and import structures. If you prefer to watch a video tutorial, check out our Getting Going with Maestro video series.

Understanding Maestro projects

All work in Maestro is organized into projects, which store the state of your Workspace, all Entries, and associated Project Table data. The project is stored on disk as a directory with a `.prj` ending, but to any non-Schrödinger program behaves like any other directory, so you can e.g. look into the project with a file manager and create or edit files therein. There are two types of projects: scratch projects and named projects.

Named projects

You can create a named project using **File → New Project** or by saving a scratch project using **File → Save Project As**. Once a project is saved as a named project, all work you do in the project is automatically saved to disk.

If you want to share a project with your colleagues, you can save it as a read-only project archive (`.prjzip`) file by going to **File → Save Project As...**

TIP: We recommend avoiding non-ASCII characters and spaces in file and project names. Use dashes or underscores instead.

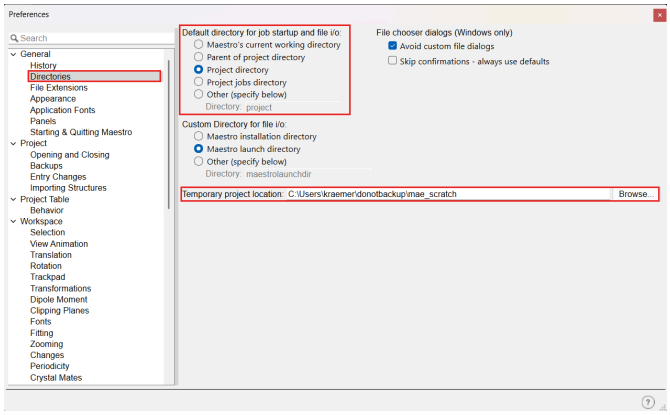
WARNING: Due to technical limitations, OneDrive folders should not be used to store Maestro projects.

Scratch projects

When you start Maestro without opening a named project, or after opening a `.prjzip`, a temporary scratch project is created. Any work you do in a scratch project is lost when you close Maestro unless you save the project as a named project (by going to **File → Save Project As**). You can customize the location on disk where scratch projects are stored while the associated Maestro session is active in **Preferences → General → Directories → Temporary project location**.

The Working Directory

Any files you generate in a Maestro session (e.g. from running jobs) will be written to the current Working Directory. This directory is also where all file chooser dialogs will initially open. The default Working Directory that Maestro uses out of the box depends on your operating system, but you can (and should) tweak the Preferences to match how you want Maestro to behave.



TIP: Go to Preferences → General → Directories → Default directory for job startup and file i/o to set your preference for the default Working Directory from the available options. We recommend choosing **Project directory** as the default Working Directory, making job results easy to find and reducing the likelihood of name clashes with jobs from different projects. Should you want to use a specific Working Directory for a project, you can use **File → Change Working Directory** to override the preference for a given Maestro session.

TIP: You can have Maestro show the current working directory and/or full project path in its title bar to avoid confusion when working with multiple Maestro sessions at a time (**Preferences → General → Appearance**).

Housekeeping within projects

Entries are the building blocks of a Maestro project. An entry consists of zero or more molecules (structures) grouped together and assigned a title. See Project Entries for more information. Entries can have properties that are related to the entire collection of atoms (entry properties), and properties for specific atoms (atom properties). Entries can be collected into entry groups, and groups can be nested within other groups for housekeeping.

Learn more about Entries here.

TIP: Entries can also store information on jobs that were run to generate structures or data and files associated with structures or data. Some jobs add shortcuts to analysis panels or common next tasks to their output entries in the form of a small icon on the right side of the Entry's row. For more in-depth operations on the data associated with a given Entry, you can use the Project Table.

You can import structures and their properties from a wide variety of other file formats, including PDB, MDL SD, Sybyl Mol2, ChemDraw, and MOE formats, native Maestro files and many others supported by the OpenBabel file conversion utility. You can also fetch structures directly from the Protein Data Bank via **File → Get PDB**.

Learn more about importing structures here.

TIP: You can copy and paste molecular data (including e.g. SMILES strings or ChemDraw structures) in your clipboard into Maestro to create new entries.

Navigating the Maestro Workspace with keyboard and mouse

TIP: We recommend using Maestro with a three-button mouse with a scroll wheel. You can change between common control presets (including one matching PyMOL) or adjust the control scheme to match your preferences from **Workspace** → **Customize Mouse Actions** .

Common mouse actions for interacting with structures in the Workspace



Maestro keyboard shortcuts and mouse controls

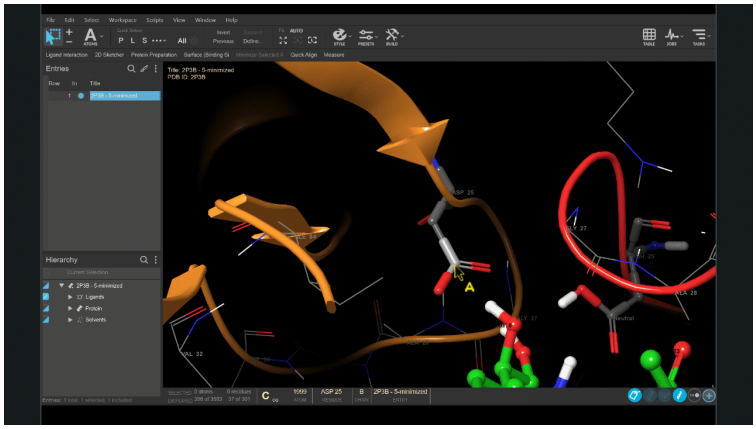


TIP: You can see and customize keyboard shortcuts in **Preferences → Shortcuts**. In addition to modifying or creating shortcuts to items in Maestro's menu bar, you can record macros of your commonly-used workflows and bind them to function keys. More details here.

Workspace selections

Maestro works on a select-first paradigm. This means that if you want to change the look of a molecule in the Workspace or you want to modify part of a structure, you first need to select those atoms within the Workspace. There are several ways to select atoms in the workspace. The two main approaches are:

With the mouse | Using selection sets



Left-click to select parts of the structure in the workspace. The selection scope for a single click is controlled with the select by type button (or the corresponding keyboard shortcuts). Double- and multiple clicks select increasingly larger parts of the structure. Shift and Ctrl work as expected. Click and drag can make rectangular or lasso selections.

TIP: All selected atoms are highlighted with a blue outline. Selected atoms that are not currently shown in the workspace are shown as hollow blue outlines.

What's next?

We recommend running through our Introduction to Structure Preparation and Visualization and A Chemist's Guide to Maestro tutorials for guided examples of basic structure visualization and editing workflows in Maestro. You can also browse our tutorial library for step-by-step examples covering various workflows. If you are unsure what resources are available for a given scientific workflow, explore our learning paths or online certification courses.

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